CCC-C-437b
NOVEMBER 8, 1960
SUPERSEDING
Fed. Spec. CCC-C-437a
May 28, 1958

FEDERAL SPECIFICATION

CLOTH, COTTON, BROADCLOTH, MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the requirements for cotton broadcloth.

1.2 Classification.

1.2.1 *Types and classes.* — The cloth shall be of the following types and classes, as specified (see 6.2):

Type I.—148 by 74, 3.5 ounces per square yard.
Type II.—140 by 56, 3.2 ounces per square yard.
Type III.—132 by 64, 3.2 ounces per square yard.
Type IV. — 116 by 56, 3.2 ounces per square yard.
Type V.—102 by 56, 3.2 ounces per square yard.
Type VI. — 82 by 56, 2.8 ounces per square yard.
Class 1.—Bleached.
Class 2.—Unbleached.
Class 3.—Dyed.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

V-T-276 — Thread, Cotton.
UU-P-268 — Paper, Kraft, Wrapping.
CCC-T-191 — Textile Test Methods.
PPP-P-51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4 — Glossary of Fabric Imperfections.
Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations of the Textile Fiber Products Identification Act effective March 3, 1960.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Standard sample.—When a standard sample is available, the cloth shall match the sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the sample is referenced. (See 6.3.)

3.2 Material.

3.2.1 *Yarn.* — The yarn shall be made from cotton which has been cleaned, carded, or combed as specified in table I, drawn and spun into single yarn (see 6.7). The contractor shall submit a certificate of compliance with each lot, certifying that combed yarns were used in the manufacturing of the cloth in types I, II, and III only.

3.3 Color.—The color shall be as specified (see 6.2). The use of sulfur dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. Presence of sulfur shall be determined as specified in 4.4.

3.3.1 *Matching.* — The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* — The dyed cloth shall show colorfastness to laundering, bleaching, perspiration, crocking, and light equal to or better than the standard sample. When a standard sample is not available or not designated as applicable to colorfastness, the dyed cloth shall show good colorfastness to laundering, bleaching, perspiration, crocking, and light. Tests shall be made as specified in 4.4.

3.4 Physical requirements.—The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

3.4.1 *Width.* — The width shall be as specified and shall be the minimum acceptable width exclusive of selvages (see 6.2).

3.5 Finish.—The cloth shall be singed, desized, bleached (class 1 only), mercerized, and shall have a clear lustrous finish. Residual sizing, finishing, or other nonfibrous material shall not exceed two percent when tested as specified in 4.4.

3.6 Shrinkage. — The cloth shall be preshrunk and shall not shrink more than one percent in either the direction of the warp or of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.7 Seam efficiency. — The finished cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.4.

3.8 Put-up. — Unless otherwise specified, the cloth shall be not less than 50 continuous yards, put up in folded pieces as specified in Federal Specification PPP–P–51 (see 6.2).

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TABLE I.—*Physical requirements*

Type	Yarn	Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (grab method) (minimum)		Weave
			Warp	Filling	Warp	Filling	
		Ounces			*Pounds*	*Pounds*	
I	Combed	3.5	148	74	74	28	Plain
II	Combed	3.2	140	56	65	25	Plain
III	Combed	3.2	132	64	75	30	Plain
IV	Carded	3.2	116	56	60	25	Plain
V	Carded	3.2	102	56	55	25	Plain
VI	Carded	2.8	82	56	30	25	Plain

3.9 Tickets. — Each piece shall have an identification ticket attached as specified in Federal Specification PPP–P–51.

3.10 Fiber identification.—Each piece of broadcloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act effective March 3, 1960.

3.11 Workmanship.—The cloth shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. —Unless otherwise specified the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Goverment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination of the end item.—Except when otherwise specified by Civilian Government agencies, examination shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated. Defects found during this examination shall be classified in accordance with 4.2.1, 4.2.2, and 4.2.3.

4.2.1 *Yard-by-yard examination.* — The required yardage of each piece shall be inspected on the face and the defects classified in accordance with the classification of defects below. The defects listed herein shall be counted regardless of their proximity to one another, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one yard. The acceptable quality levels for this examination shall be 2.5 major and 10.0 total defects (major and minor combined) per 100 units (yards). The sample size shall be based upon inspection level III, or Military Standard MIL–STD–105. The lot size shall be expressed in units of one yard each. The number of pieces from

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which the sample is to be selected (not more than one piece shall be selected from each container) shall be in accordance with table II. An approximate equal number of yards shall be examined from each piece in the sample.

Glossary Nos. Fed. Std. No. 4	Defects	Classification	
		Major	Minor
90, 91, 92	Hole, cut or tear, three or more yarns ruptured at adjoining points	X	
78, 79, 89	Smash, bruise, tender or weak spot—any	X	
11	Broken or missing yarns:		
	—Two or more contiguous, more than ½ inch missing	X	
	—Single, 2 inches or more missing		X
9	Fine filling bar or light place	X	
8, 25	Fine, uneven or light yarns—clearly visible*		X
1, 2, 26	Coarse filling bar or heavy place	X	
	Coarse yarns, clearly visible*		X
5, 6, 27, 77	Kink, slub or gout:		
	—Over 1/16 inch in thickness	X	
	—1/16 inch or less in thickness, clearly visible*		X
7, 45	Surface knot or fuzz ball—more than ⅛ inch in diameter		X
3, 5	Jerked in filling, slough off:		
	—Over 1/16 inch in thickness, regardless of length	X	
	—1/16 inch or less in thickness and more than 9 inches in length	X	
	1/16 inch or less in thickness, but not over 9 inches in length—clearly visible*		X
17, 35	Open place, or hitchback:		
	—Resulting in a weak spot	X	
	—Not resulting in a weak spot, clearly visible*		X
79	Abrasion mark—clearly visible* fuzziness but not resulting in a weak spot		X
4, 10	Mispick, two or more picks in same shed	X	
13	Floats or harness skips:		
	—Multiple ¼ inch or more in combined warp and filling directions	X	
	—Multiple less than ¼ inch in combined warp and filling directions		X
	—Single, two inches or more		X
16, 37, 38, 40, 41	Tight or slack yarn, slack or tight twist yarn, wrong draw, reed mark—clearly visible*		X
42, 76	Temple mark—clearly visible*		X
50	Selvage tight—resulting in pronounced slackness or waviness in cloth		X
51	Selvage scalloped—indentation exceeding ½ inch		X

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Glossary Nos. Fed. Std. No. 4	Defects	Classification	
		Major	Minor
52	Selvage slack—resulting in pronounced waviness		X
14, 57	Shade bar, mixed filling or finishing mark	X	
74, 75, 76, 69, 70**	Spot, stain or streak—clearly visible*	X	

* Clearly visible at normal inspection distance (approximately 3 feet).
** Except defects of spot, stain or streak, shall be counted regardles of cause, i.e., not restricted to rust, oil or grease.

4.2.2 *Over-all examination.* — Each defect listed as follows shall be counted no more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the number of defects acceptable shall be as shown in table II.

Defects

Width less than specified.

Uneven weaving, clearly visible.

Baggy, ridgy, or wavy cloth — clearly visible, will not flatten under normal pressure.

Selvage curled, folded, rolled or slack—extending intermittently or continuously over more than half the length of the piece.

Off shade, shaded side to side, side to center, end to end, or throughout the piece.

Mottled, cloudy or poor dye penetration.

Objectionable odor, overall uncleanness.

4.2.3 *Examination for length.*

4.2.3.1 *Examination for length of individual pieces.* — The pieces shall be examined for gross length. Any gross length found to be less than the minimum specified, or any gross length found to be more than two yards below the gross length marked on the piece ticket, shall be considered as a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as sample (sample size) for this examination and the number of defects acceptable shall be in accordance with table II.

TABLE II.—*Sample size*

Lot size in yards	Sample size in pieces	Acceptance number
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 to 550,000	25	0
550,001 and over	50	0

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.3.2 *Total yardage in sample.* — The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets. The pieces examined shall be those selected for examination of individual pieces.

4.3 Examination of preparation for delivery.—An examination will be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects will be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection

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lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Marking ------- (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials ------	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship --	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight --------	Gross weight exceeds requirements.

4.4 **Tests.**—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table III herein shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 4½ continuous yards, full width of the finished fabric. The lot shall be unacceptable if one or more units fails to meet any requirement specified. For each characteristic, the sample size shall be in accordance with the following:

Lot size in yards	Sample size
Less than 800 ----------	2
801 to 22,000 ----------	3
22,001 and over ----------	5

4.4.1 *Test for presence of sulfur.* — Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. To preclude interference in dye identification, sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process. One to 1.5 grams of the finished material, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of 5 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 cc. of distilled water; add a drop of glacial acetic acid to clear up if cloudy), over the mouth of the test tube. (Punch needle hole in paper.) Heat slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur. One determination shall be made from each unit of product and results reported as "pass or fail".

5. PREPARATION FOR DELIVERY

Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.

5.1 **Packaging.**—Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Military agencies.*

5.1.1.1 *Levels A, B, and C.* — The cloth, flat folded as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.1.2 *Civil agencies.*

5.1.2.1 *Levels A, B, and C.* — The cloth, flat folded as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51 except material purchased in bolt form shall be completely enclosed in a wrap of 30-pound basis weight kraft paper conforming to grade B of Federal Specification

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TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method
Weave[1]	Table I	Visual
Yarns per inch	Table I	5050
Weight	Table I	5041
Breaking strength	Table I	5100
Colorfastness to:		
Light[2]	3.3.2	5660
Chlorine bleaching	3.3.2	5600
Perspiration[3]	3.3.2	5680
Crocking	3.3.2	5651
Laundering	3.3.2	5610
Presence of sulfur	3.3	(See 4.4.1)
Nonfibrous materials[4]	3.5	2611
Mercerization	3.5	[5]
Shrinkage[6]	3.6	5550
Seam efficiency[7]	3.7	5110

[1] One determination per unit of product reported as "pass or fail".

[2] In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] The transfer cloth shall consist of a warp of polyester fiber or Orlon 81, and a 6-fiber filling of equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

[4] The results shall be calculated from the following formula:

$$\text{Nonfibrous content, percent} = \frac{O - S}{O} \times 100$$

[5] Acceptance with respect to this characteristic may be based on a contractor's certification of compliance (see 3.5).

[6] The shrinkage of the specimen shall be calculated as follows and reported to nearest 0.1 percent:

$$\text{Shrinkage, percent} = \frac{A - B}{A} \times 100$$

where:

A = Average of initial measurements (3 specimens)

B = Average of measurements after laundering (3 specimens)

The 3 specimens shall be selected from the cloth (unit of product) as follows: One specimen from each side of the cloth within 3 inches of the selvage and 1 specimen from the center of the cloth. No 2 specimens shall contain the same filling yarns.

[7] The needle shall be 0.032 ± 0.001 inch across the blade at the eye and shall have a round point. The thread shall be type IA1 conforming to Federal Specification V–T–276; the needle thread shall be ticket No. 60, 3-ply and the looper thread shall be ticket No. 70, 2-ply.

UU–P–268. The wrapping shall encircle the folded fabric at least 1½ times and shall be sufficiently wide to fold over the selvages. The wrap shall be secured with gummed paper tape. Each piece shall be wrapped so as to have the attached identification ticket clearly exposed on the outside of the package.

5.2 Packing.—Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 *Military agencies.*

5.2.1.1 *Levels A, B, and C.* — The cloth shall be packed in accordance with the applicable requirements of Federal Specification tion PPP–P–51.

5.2.2 *Civil agencies.*

5.2.2.1 *Levels A, B, and C.* — The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51 except that for civilian agencies four paper-wrapped pieces (packages) 5.1.2.1, shall be packed flat one in length, one in width, and four in depth within a shipping container.

5.3 Marking.—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use.—The cloth covered by this specification is intended specifically for men's and women's shirts.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

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(*a*) Title, number, and date of this specification.
(*b*) Types and classes of cloth required (see 1.2.1).
(*c*) Length of cloth if other than specified (see 3.8).
(*d*) Color of cloth required (see 3.3).
(*e*) Width of cloth required (see 3.4.1).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(*g*) Use of Military Standard MIL–STD–105 (see 4.2).

6.2.1 In the preparation of contracts or orders, it should be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of this cloth.

6.3 **Standard sample.** — For access to a standard sample, address the procuring office issuing the invitation for bids.

6.4 **Commercial grey designation.**—It is suggested that cloth of the following commercial grey designations (or prorata) be used to produce cloth meeting the requirements of this specification.

Commercial grey designation (suggested)

Type	Width	Yarns per inch	Weight in yards per pound
I	37½	144 x 76	4.15
II	37	136 x 60	4.00
III	37	128 x 68	4.00/4.25
IV	37	112 x 60	3.85
V	37	100 x 58	4.00
VI	40	80 x 60	4.50

6.5 Level B packaging and packing shall be used by ordering agencies only when shipments are known to be made to a holding Supply Depot or Warehouse from which partial shipments will be made to ultimate consumers.

6.6 **Transportation description.** — The transportation description applicable to this item is:

Cotton piece goods, in the original piece, finished, not otherwise indexed by name.
Carload minimum weight, any quantity.
Motor, volume minimum weight, any quantity.

6.7 **Procurement by civilian agencies.** — Broadcloth procured by civilian Government agencies under this specification may contain yarns in excess of single ply providing the broadcloth conforms to all other requirements of this specification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M Q
Navy—S MC
Air Force—WADC.

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